

NOISY-PATCH CLASSIFICATION FAVORS TOKEN TRANSFORMATION OVER DESTRUCTIVE FORGETTING

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ABSTRACT

Reversible ViTs are constrained by a layerwise bijectivity requirement. This means that they cannot destructively forget information from layer to layer. One may hypothesize that such a constraint would lead to worsened performance on tasks such as noisy patch classification. However, we see no difference in performance versus a Plain ViT, even with constrained model dimension. We use a series of probes, causal interventions, a gated residual stream update variant of the ViT and comparisons to a long short-term memory model, and find no evidence that forgetting is beneficial to noisy patch classification performance. Instead, models perform best when they learn to transform noise patch tokens into something that resembles a good guess of what was covered. We conclude that the Rev model’s matching of plain performance stems from this lack of a need for destructive information loss.

1 INTRODUCTION

The reversible residual network framework, as defined by Gomez et al. (2017), is a powerful method by which one can effectively make memory scaling constant in backpropagation during training. This is possible since with reversibility, one can reconstruct module inputs from outputs and no longer have to store intermediate activations in memory while training. The tradeoff is that we now recompute activations, which takes longer. Kitaev et al. (2020) introduce the Reformer, which extends the reversible framework to transformers. This idea is further developed for vision transformers by Mangalam et al. (2022).

It is important to note that contrary to a reversible recurrent neural network as explored by MacKay et al. (2018) where reversibility applies on a timestep basis, transformers apply reversibility on a layer basis so memory requirements no longer scale with network depth. This is still a powerful optimization as many popular vision transformer architectures today such as large BEiT variants (Bao et al., 2022) have dozens of layers, leading to 10-20x reductions in peak GPU memory usage during training.

Although reversible transformers seem to be a great alternative for more hardware constrained environments, these networks must be bijective from layer to layer. This is to say that no information can be destroyed between layers since enough must be retained to reconstruct the input. Such a constraint on the model raises the question about its performance on certain tasks. Consider the task of image classification on images with noise patches covering some portion of the full frame (Figure A1). Intuitively a model that can suppress and destroy distractors like the noise can be hypothesized to outperform a model that cannot, especially in constrained hidden dimension regimes where the model may not have enough state capacity to fully represent all patches without discarding the noise. However, experimental results show that this expectation is not met: the Rev ViT matches plain performance throughout all model dimensions (Figure 1a) tested.

In this paper, we focus on why the Rev ViT does not under-perform its plain counterpart on the noisy patch classification task. We run probes including measuring layerwise reconstruction ability, layerwise classification accuracy, layerwise AUROC for determining whether a particular patch representation originated from a noise token or not and several causal intervention probes. We also look at how internal representations differ in fine-tuned reconstruction models for noisy patch classification, gated residual stream variants of the ViT, and also long short-term memory (LSTM) models

to answer this question. We conclude that the task of noisy image classification does not benefit from destructive information loss, and performance actually degrades with such behavior. What is important is the ability to transform internal representations of noise tokens into something that resembles a close approximation of what the noise has replaced. That information loss hurts and strong learned transformations succeed may explain why the Rev model, which has no trouble learning such transformations, matches the performance of the Plain ViT, which learns not to destructively forget.

2 RELATED WORK

Reversible architectures. Reversible networks reduce activation memory by reconstructing intermediate states during backpropagation rather than storing all activations. This idea has been developed in reversible residual networks (Gomez et al., 2017) and later extended to transformers through the Reformer architecture (Kitaev et al., 2020). Reversible vision transformers apply this principle to visual recognition (Mangalam et al., 2022). Our work differs from these memory-efficiency studies by asking whether the bijective layer constraint changes what information a vision model can usefully discard or transform when inputs contain corrupted patches. To the best of our knowledge, such a study on the limitations of reversible transformers has not yet been conducted.

Masked image modeling and reconstruction. Masked image modeling methods train vision models to infer missing image content from visible context, such as in masked autoencoders (He et al., 2022). This line of work is relevant because noisy-patch classification can be interpreted as requiring the model to reason about corrupted or occluded patches. However, our focus is not on using reconstruction as a pretraining objective itself, but on using reconstruction-trained models and representation probes to understand whether classification benefits from deleting corrupted tokens or transforming them into surrogate representations.

Forgetting and gating. Gated recurrent models such as forget-gated LSTMs (Hochreiter & Schmidhuber, 1997; Gers et al., 2000) provide an explicit mechanism for controlling which information is written, retained, or forgotten. This makes them a useful comparison point for testing whether noisy-patch classification requires selective suppression of corrupted inputs. Our experiments compare this explicit gating mechanism with Rev and Plain ViTs, and find that neither the transformer nor the LSTM evidence supports destructive forgetting as the useful strategy.

3 REV AND PLAIN ViTs STORE NOISE TOKEN INFORMATION

Reversible Architecture The core architecture of a Reformer-style reversible transformer layer splits the residual stream into two streams, x_1 and x_2 . One stream is updated using attention, while the other is updated using the feedforward block:

$$y_1 = x_1 + \text{Attn}(x_2),$$

$$y_2 = x_2 + \text{FFN}(y_1).$$

Because the updates are additive, the layer can be inverted exactly by reversing the two updates:

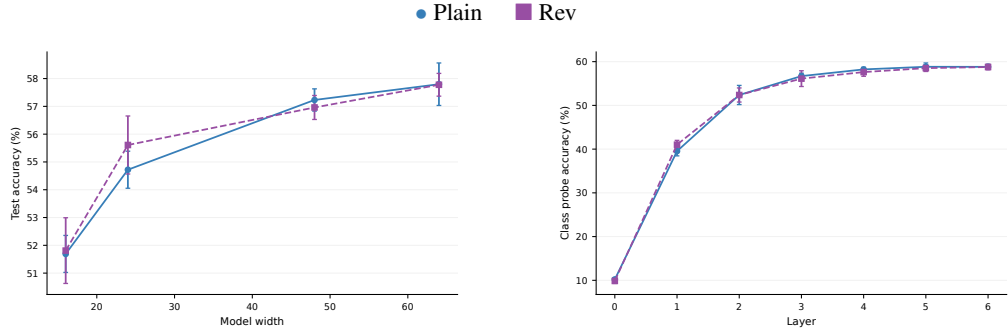
$$x_2 = y_2 - \text{FFN}(y_1),$$

$$x_1 = y_1 - \text{Attn}(x_2).$$

Thus, the full residual state (x_1, x_2) is preserved across depth even though each sublayer only operates on one stream at a time (Kitaev et al., 2020).

Experimental Setup In this work, we match the operator dimension at 64, so each attention/MLP block acts in the same width. The Rev model has a doubled residual state because of the split stream, resulting in a very slight 2.87% difference in parameters (Table A4).

We use the CIFAR-10 dataset in our experiments. Each image is split into non-overlapping 4×4 patches. For the noisy-patch classification task, we choose a noise patch fraction which represents what percent of the total 64 patches are noise patches. For each image, this fraction of patch locations is sampled uniformly at random without replacement. The selected patches are then replaced



(a) Width sweep on CIFAR-10 with noisy patches.

(b) Layerwise class-probe accuracy for Scratch Plain and Scratch Rev ViTs at 25% noise.

Figure 1: Scratch Plain and Scratch Rev ViTs show similar performance across model width and similar layerwise class decodability.

entirely with independent uniform random pixel values, while all unselected patches remain unchanged. Thus, the model receives an image containing a mixture of clean CIFAR-10 patches and localized noise patches, together with the original image-level class label. Each raw patch contains $3 \times 4 \times 4 = 48$ values. A learned linear layer maps each patch to a 64-dimensional patch embedding. A learned `cls_token`, also 64-dimensional before projection, is added. Then the model concatenates learned positional embeddings: each token receives a 32-dimensional learned position vector. We concatenate instead of adding since this allows for better reversible performance with no adverse effect on the Plain ViT. Thus, each token temporarily has $64 + 32 = 96$ dimensions, after which a learned `input_proj` maps it down to the transformer input dimension of 64. Our exact experimental hyperparameters are shown in Table A4. We refer to the baseline plain and Rev ViT models as the “scratch” models in this paper, indicating that they are trained for classification from scratch.

Layerwise Accuracy Probe As shown in Figure 1a the Rev model matches the Plain ViT through all measured model dimensions. We now hypothesize that both models retain some noise token information. To see this, we conduct a series of linear probes on the models, starting with a layerwise classification accuracy probe. Both models show a sharp increase in accuracy in the first few layers and then taper off towards their final accuracies. They show no significant difference at any layer. This, as well as all other experiments, was run over three seeds to reduce variance (Figure 1b).

Layerwise Reconstruction Probe We then run a layerwise reconstruction probe. Here, we do observe a difference: the Rev model has a lower reconstruction error than the Plain model (Figure 2a). Both models start at around the same mean squared error (MSE) — 0.0148 for plain and 0.0141 for reversible. As we probe deeper layers, we see that internal representations for both models become less linearly decodable, however the Plain model is far less linearly decodable by the end than the Rev model — 0.0405 for plain and 0.0311 for reversible. We infer from this that the Rev model stores more linearly accessible input information. This is not to say that the extra information stored is a constraint, as we see performance is still matched.

Layerwise AUROC Probe The next probe answers the question: given a token’s hidden representation at layer ℓ can a simple probe tell whether that token originally came from a noisy/corrupted patch or a clean/visible patch? We achieve this by first labeling each patch token as either a 1 for noise tokens or a 0 for clean tokens. We then train a binary linear probe to classify each token at layer ℓ as either noisy or clean. The metric we use to measure the result here is the area under ROC (AUROC) which is defined as

$$\text{AUROC}_\ell = P(s(h_{\text{noise},\ell}) > s(h_{\text{clean},\ell})),$$

where $s(h)$ is the probe’s score that a token came from a noisy patch. An AUROC score of 0.5 indicates that the patches cannot be distinguished via a linear probe. What we see in the final layer is that for both Plain and Rev, the AUROC score is quite high at 0.849 while reversible is higher

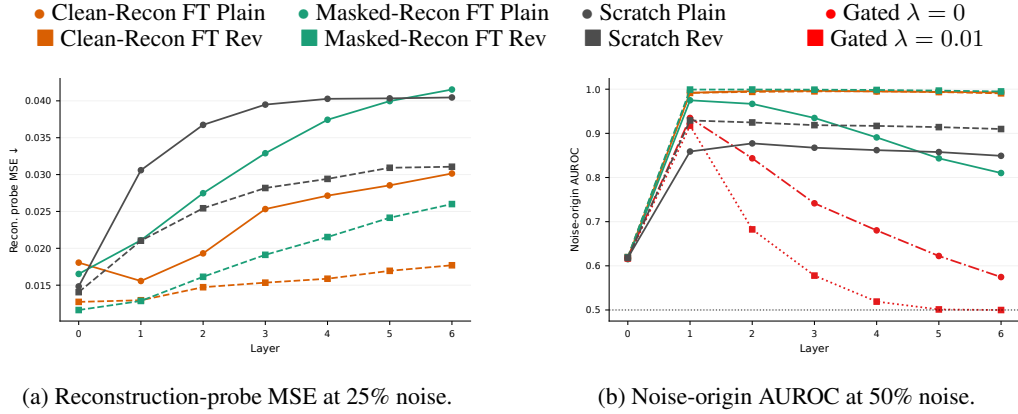


Figure 2: Layerwise probe results. Left: reconstruction-probe MSE. Right: noise-origin AUROC.



Figure 3: Layer 2 input reconstruction probe visualization.

at 0.910 (Figure 2b). Thus, both models retain enough information about the type of patch each token originated from such that we can linearly decode this information in deeper layers. This high AUROC score is sustained through all layers after the very start. This is evidence that both models, but most significantly the Plain model, are not discarding all noise-token specific information.

This lack of destructive forgetting that seems to be apparent in even the Plain model, which has the ability to delete, is more obvious when we inspect the layerwise reconstructions of these models. We train a linear probe at each layer to reconstruct the model input and we observe that both the plain and Rev models learn to make the noise patches look more like clean tokens and make the clean patches look more like noise tokens (Figure 3). Further analysis of this behavior will follow later on in the paper, but it is apparent visually that neither model seems to be destroying noise information or discarding noise patches.

4 RECONSTRUCTION TRAINING IMPROVES CLASSIFICATION BY IMPROVING TOKEN SUBSTITUTION

We compare two reconstruction-pretrained ViT variants: a Masked-Recon ViT and a Clean-Recon ViT. Both use the same Plain/Rev ViT architecture as the Scratch classification baselines, but are first trained with a patch reconstruction objective before being fine-tuned for noisy-image classification. The Masked-Recon model is pretrained on noisy inputs and learns to reconstruct the original clean

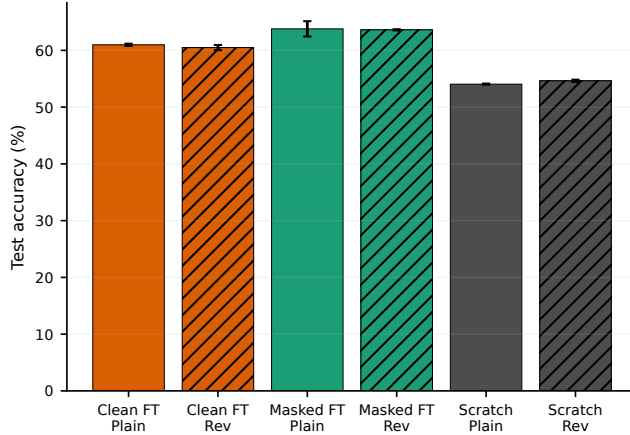


Figure 4: Main CIFAR-10 performance comparison at 50% noise.

content of the corrupted patches, with the reconstruction loss applied only to those noisy patches. The Clean-Recon model is pretrained on clean images and learns to reconstruct all image patches. After this reconstruction pretraining stage, we replace/use the classification head and fine-tune each model on the noisy CIFAR classification task. Both reconstruction-pretrained models substantially improve classification accuracy over Scratch baselines (Figure 4), with Masked-Recon pretraining giving the strongest performance at intermediate noise-patch frequencies.

We run the same probes as with the Scratch models on the fine-tuned models as well. For reconstruction, the Clean-Recon FT models generally have the lowest reconstruction error, consistent with their pretraining objective of reconstructing input patches. The Masked-Recon FT Plain ViT, however, performs at least as bad as the Scratch Plain model. This makes sense in context with visual outputs of the layerwise reconstructions (Figure 3). However, in all cases, we do not see a significant increase in reconstruction error, which signifies that this performance boost from reconstruction pretraining has little to do with learning to forget.

As discussed prior, the Scratch models have a sort of leaking effect: noise tokens look more like clean guesses while clean tokens look more like the average noise token. The Clean-Recon FT model, however, seems to quarantine its noise patch representations to exactly those tokens while the Masked-Recon FT model attempts to guess what the mask is covering. These qualitative reconstructions suggest that reconstruction pretraining improves performance by changing how corrupted tokens are represented. Rather than spreading noise into clean-token representations, the pretrained models either quarantine corrupted tokens or transform them toward plausible clean-token surrogates.

Reconstruction-pretrained models also generally show substantially higher AUROC probe scores. The Masked-Recon FT model is a slight exception in that it also initially has higher AUROC scores, but these soon drop to slightly below (albeit still high) Scratch model levels. This is to say that the reconstruction process may have given the models a better understanding of what tokens come from clean versus noise patches and this information may be crucial in the models’ ability to selectively transform noise and clean tokens differently. We see the Masked-Recon FT model able to transform noise patches into clean guesses and we see the Clean-Recon FT model able to quarantine noise patches from the clean patches.

5 CAUSAL INTERVENTIONS FAVOR SUBSTITUTION OVER DELETION

We have established a correlation between learned transformations of noise tokens and high performance. We now aim to strengthen this observation with causal evidence. We perform layerwise interventions in which the internal representation of each noisy patch token is replaced with one of

Table 1: Layer-0 causal replacement intervention on CIFAR-10 with 25% noise. Baseline accuracy is reported in percent. Intervention columns report change in classification accuracy, in percentage points, relative to the no-intervention baseline. The $\pm$ values show 95% confidence intervals across seeds.

Model	Baseline acc.	Oracle clean	Mean visible	Zero
Scratch Plain	57.97 ± 0.89	$+1.49 \pm 0.43$	-17.85 ± 1.44	-0.41 ± 0.56
Scratch Rev	57.87 ± 0.33	$+2.54 \pm 0.79$	-14.35 ± 3.47	-1.31 ± 0.42
Masked-Recon FT Plain	68.93 ± 1.76	$+2.11 \pm 0.58$	-8.02 ± 1.24	-3.87 ± 2.56
Masked-Recon FT Rev	67.81 ± 0.45	$+2.11 \pm 1.09$	-8.20 ± 5.45	-2.01 ± 0.54

three alternatives: a clean-oracle representation, an average visible-token representation, or a zero vector. At layer index l , replacement is applied to the stored representation after layer l , and the model continues from layer $l + 1$. For layer 0, this means the intervention is applied to the embedded input state before the first transformer block. In all interventions, only noisy patch tokens are replaced; clean patch tokens and the class token are left unchanged.

Clean Oracle Intervention The clean oracle is obtained by running the same model on the corresponding clean input image and extracting the matching patch-token representation at the same layer. Introducing the clean-oracle representation in early layers increases accuracy by roughly two percentage points across both scratch and Masked-Recon FT models.

Zero Vector Intervention For the zero-vector intervention, we replace each noisy token’s internal representation with zeros in that layer’s hidden-state space. This lowers accuracy by up to 3.87 percentage points in the worst case, for the Masked-Recon FT Plain ViT.

Average Token Intervention The average-token intervention replaces each noisy token with the mean of the visible, uncorrupted patch-token representations from the same image at the same layer, excluding the class token. This mean is computed from the noisy-input forward pass. This intervention performs worst: Scratch models show up to a 17.85 percentage point decrease in accuracy, while Masked-Recon FT models show a roughly 8 percentage point decrease (Table 1).

For Rev models, the intervention is defined analogously, except that the replaced state is the concatenation of the two reversible streams.

The results indicate that in the ideal case, internal representations match those of the clean state. Deleting noise information explicitly sees a decline in performance but more importantly the mean-visible replacement is especially harmful, suggesting that poor surrogate tokens are worse than simple deletion. This may indicate why the Masked-Recon FT model performs the best: it is optimized to make good guesses for noise patches.

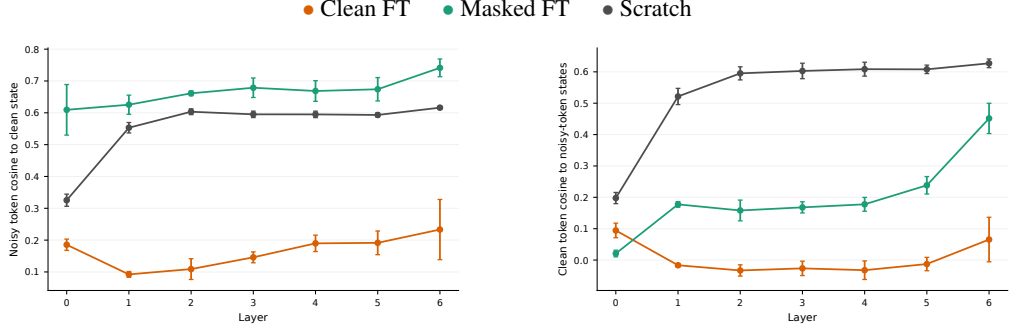
6 SIMILARITY ANALYSES SUPPORT THE SURROGATE-TOKEN MECHANISM

Cosine Similarity to Clean Oracle That a clean representation is most beneficial is complemented by a probe that measures layerwise cosine similarity between noise token representations and the corresponding clean oracle representation (Figure 5a). We see that over layers there is a gradual increase in cosine similarity, indicating that all models see this as important to the overall noisy-patch classification objective.

Cosine Similarity to Noise Tokens We also conduct a probe measuring cosine similarity between all pairwise clean tokens and existing noise tokens in that sample:

$$\frac{1}{|C||N|} \sum_{i \in C} \sum_{j \in N} \cos(h_{i,\ell}^{\text{clean}}, h_{j,\ell}^{\text{noise}}).$$

We see that the Scratch model’s clean tokens are substantially closer to the noise tokens than those in the fine-tuned models, especially in early and intermediate layers (Figure 5b). The Clean-Recon FT



(a) Noisy-token cosine similarity to clean-run token states. (b) Clean-noise token similarity within the corrupted run.

Figure 5: Similarity probes for the Plain ViT at 50% noise. Left: noisy-token states are compared to their corresponding clean-run oracle states. Right: clean-token states are compared to existing noisy-token states from the same corrupted-input run.

model has the lowest similarity, which supports our observation that this model seems to quarantine noise into the patches that input it. This indicates that it is not only important that noise token representations are transformed into close approximations of the underlying clean tokens, it is also important that existing clean tokens retain their clarity through the network, something that the Scratch models struggle with. The AUROC scores indicated that the fine-tuned models may have a better way of representing clean versus noisy and this internal understanding may contribute to the ability to prevent this noise leakage that is prevalent in the Scratch models.

Attention to Noise Tokens We additionally probe the attention mass that the clean tokens give to noise tokens across these models to see if that could explain why the Scratch models have more leakage. We observe that the Scratch models’ clean tokens typically allot slightly more attention weight to the noise tokens than in Masked-Recon FT, especially in intermediate layers. We observe the effect is more pronounced in the Rev models (Table A1). We see a similar difference in the attention weight that the `cls_token` gives to noise tokens (Table A2).

7 DESTRUCTIVE NOISE SUPPRESSION HURTS PERFORMANCE

Residual Update Gating We aim to rule out the possibility that forgetting or destructively suppressing noise token representations can improve performance. Our first approach in showing this is defining a new ViT variant with explicit gating for residual updates. Given an input stream X_ℓ and a candidate update $\tilde{X}_\ell$, the gated update is

$$X_{\ell+1} = g_\ell \odot X_\ell + (1 - g_\ell) \odot \tilde{X}_\ell. \quad (1)$$

We apply one scalar per token, thus, when $g_{\ell,i} \approx 1$, token i mostly retains its previous state, while when $g_{\ell,i} \approx 0$, token i is mostly overwritten by the candidate state. In the case of the transformer, the candidate state is computed via either the attention block or the feed-forward network (FFN). We also include a regularization term that penalizes low gate values on clean tokens. This regularizer encourages selective suppression of noisy tokens while preserving clean-token updates.

$$\mathcal{L}_{\text{gate}} = \frac{1}{LBN} \sum_{\ell=1}^L \sum_{b=1}^B \sum_{i=1}^N m_{b,i} g_{\ell,b,i}. \quad (2)$$

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{gate}} \mathcal{L}_{\text{gate}}. \quad (3)$$

We test this gated variant with and without the regularizer. With the regularizer, we test a series of λ multipliers and report results for $\lambda = 0.01$. Larger lambdas yielded worse performance. What

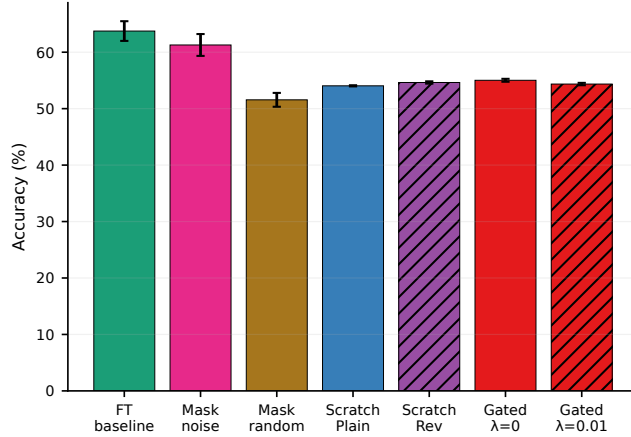


Figure 6: Suppression and attention-masking interventions on CIFAR-10 at 50% noise. The FT baseline, Mask noise, and Mask random bars are paired evaluations of the Masked-Recon FT Plain ViT. Scratch Plain and Scratch Rev are independently trained scratch baselines. Gated $\lambda = 0$ and $\lambda = 0.01$ are separately trained gated-suppression models.

we see is that there is no discernible improvement in performance between this gated variant and the Scratch models (Figure 6). However, we do see a reduction in AUROC probe scores to 0.575 for $\lambda = 0$ and 0.500 for $\lambda = 0.01$ (Figure 2b) indicating that, on at least a surface level, the gated models are retaining less information about noise tokens than the other models studied thus far. That this explicit information loss shows no improvement over baseline indicates that such an approach may not be the right inductive bias for this task.

Attention Masking We try another approach to suppress noise information: attention score masking. Here, we use pretrained classification models and, during inference, mask out attention logits belonging to noise token keys. The result is that no attention mass is placed on noise tokens. We apply this to the Masked-Recon FT model and see a reduction in model performance, once again indicating that destructive efforts of deleting noise token information does not improve results and may actually be counterproductive.

8 LSTMs DO NOT SUPPRESS NOISE INFORMATION EITHER

We extend this result from solely vision transformers to LSTMs as well, to see whether this phenomenon is model specific or something more fundamental about the noisy-patch classification task itself. The LSTM baseline processes image patches sequentially, following the broader line of work using recurrent sequence models for image classification (Tatsunami & Taki, 2022). An LSTM updates its cell state as

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t,$$

where i_t controls how much new candidate content is written and f_t controls how much previous memory is retained. To use the LSTM model for image classification, we feed in a single image patch encoding per timestep in row-major order. Since order is constant for all samples, we do not provide additional positional encodings. Once all 64 patches have been fed to the model, we give it 4 blank compute steps to further process all of its memory. Finally, we have one output step where the LSTM makes its classification. The hyperparameters for this baseline are shown in Table A3.

If the LSTM were explicitly suppressing noise patch information, we would expect to see lower input-gate activations on noisy timesteps and stronger forgetting behavior aligned with noisy patches. We therefore compare average input- and forget-gate activations on clean versus noisy patch timesteps across noise levels. The resulting gate statistics are shown in Table 2.

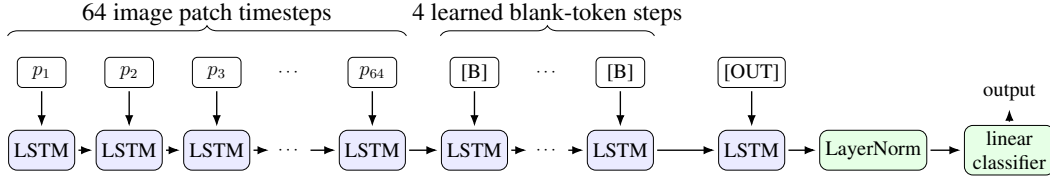


Figure 7: LSTM image-classification baseline. The image is split into 4×4 patches, each patch is linearly embedded, and the resulting 64 patch tokens are fed sequentially into the LSTM. Four learned blank compute tokens allow the recurrent state to further process the image, followed by a learned output token. A final linear head predicts the class from the normalized hidden state at the output-token timestep.

Table 2: Mean LSTM gate activations on clean and noisy patch timesteps across noise levels. Input gates are not lower on noisy timesteps, and forget gates are slightly higher on noisy timesteps, providing no evidence for selective gate-based suppression or destructive forgetting of noisy patches. The $\pm$ values show 95% confidence intervals across seeds.

Noise	i_{clean}	i_{noise}	f_{clean}	f_{noise}
25%	0.526 ± 0.025	0.524 ± 0.027	0.741 ± 0.005	0.749 ± 0.005
50%	0.500 ± 0.025	0.496 ± 0.027	0.738 ± 0.009	0.747 ± 0.010
75%	0.482 ± 0.025	0.478 ± 0.026	0.693 ± 0.020	0.703 ± 0.022
90%	0.496 ± 0.015	0.496 ± 0.017	0.632 ± 0.013	0.638 ± 0.014

As shown in Table 2, although LSTMs contain explicit gates that could, in principle, suppress noisy inputs, the measured gate activations do not show selective noise suppression. Across noise levels, input-gate activations are nearly identical across clean and noisy patch timesteps, while forget-gate activations are slightly higher on noisy timesteps. Thus, even in a gated recurrent baseline, noisy-patch classification is not solved by selectively closing input gates or increasing forgetting at noisy timesteps.

9 DISCUSSION

These experiments suggest that Rev ViTs match Plain ViTs because noisy-patch classification, in this setting, does not benefit from destructive forgetting. Although Rev ViTs cannot discard information between layers, strong Plain ViTs do not appear to rely on such deletion either.

The noise-origin AUROC probes show that both plain and Rev models retain linearly decodable information about noisy tokens. The causal replacement experiments show that replacing noisy-token representations with clean oracle representations improves accuracy, while replacing them with zeros or global average visible-token representations hurts performance. This suggests that the important policy to learn is not noise suppression, but rather noise transformation to a good representation of the underlying clean patch. The Reconstruction-fine-tuned models further support this view: Masked-Recon improves classification because it encourages the model to infer what the corrupted patches may contain.

The gated residual update variant models and the attention masking interventions provide empirical evidence that forcing explicit noise suppression offers no benefit to the classification objective and may actually reduce performance. This behavior is also exhibited by the LSTM model, which does not change its forget and input gate activations depending on whether a token came from a noise or clean patch.

These findings imply that reversibility is not inherently incompatible with noisy or corrupted inputs. A Rev model is constrained from destroying information, but it can still transform representations. When a task is best solved by transforming corrupted tokens into useful clean tokens, reversibility may impose little or no performance cost. The constraint should matter more in tasks where the optimal computation requires irreversible deletion or compression of irrelevant information, but we do not identify such a task in this study.

There are several limitations to this study. The experiments focus on CIFAR-10, patch-level synthetic noise, and relatively small ViT architectures. The results may differ for larger scale datasets, natural occlusions, structured corruptions, or tasks requiring fine-grained localization. Several other tasks were tested in this work, including a pure reconstruction task, noisy canvas classification where a clean 32×32 image is randomly placed on a larger 64×64 canvas of noise, and a shape segmentation task. In all of these other tasks, which were not the focus of this study, the Rev ViT still matches plain performance. While the current work defines a reason for this performance match in the noisy-patch classification task, it does not do so for these other tasks where the same phenomenon is observed. Future work should test whether the same surrogate-token mechanism appears in larger vision models and whether there are vision tasks where destructive forgetting is genuinely beneficial.

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A APPENDIX



Figure A1: An example input image with 25% uniform noise patches.

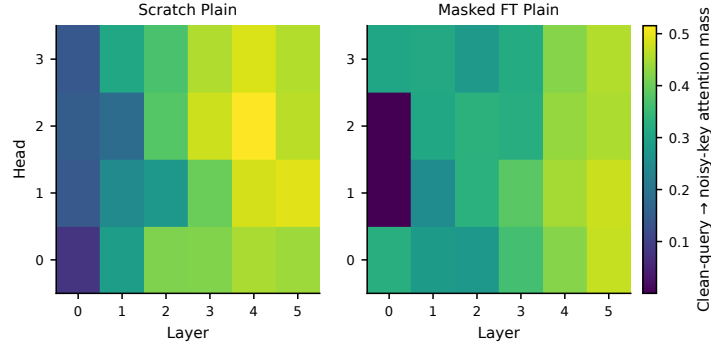


Figure A2: Head-wise attention-routing heatmaps for the Plain ViT on CIFAR-10 at 50% noise. Each panel shows mean attention mass from clean-query tokens to noisy-key tokens, with rows as heads and columns as layers. The packaged attention-routing tables were available for scratch and Masked-Recon FT checkpoints, but not for Clean-Recon FT checkpoints. Lower values indicate less routing from clean tokens into noisy-token positions.

Model	Layer	Scratch	Masked FT	Scratch – Masked FT
Plain	0	0.2001	0.2441	-0.0440
	1	0.2712	0.2741	-0.0029
	2	0.3734	0.3101	0.0633
	3	0.4325	0.3517	0.0808
	4	0.4717	0.4520	0.0197
	5	0.4771	0.4704	0.0068
Rev	0	0.1624	0.2899	-0.1275
	1	0.2991	0.1674	0.1317
	2	0.3919	0.2847	0.1072
	3	0.4198	0.3372	0.0826
	4	0.4489	0.4418	0.0072
	5	0.4465	0.4710	-0.0246

Table A1: CIFAR-10 attention mass allocated to noisy tokens by layer, averaged across heads and noise fractions.

Model	Layer	Scratch	Masked FT	Scratch – Masked FT
Plain	0	0.1917	0.2351	-0.0434
	1	0.2532	0.2623	-0.0091
	2	0.3452	0.3089	0.0363
	3	0.4048	0.3520	0.0528
	4	0.4485	0.4388	0.0097
	5	0.4469	0.3986	0.0483
Rev	0	0.1165	0.2538	-0.1373
	1	0.2816	0.1485	0.1331
	2	0.3300	0.2539	0.0761
	3	0.3885	0.3279	0.0606
	4	0.4185	0.3776	0.0409
	5	0.4259	0.4245	0.0014

Table A2: CIFAR-10 class-token attention mass allocated to noisy tokens, averaged across heads and noise fractions.

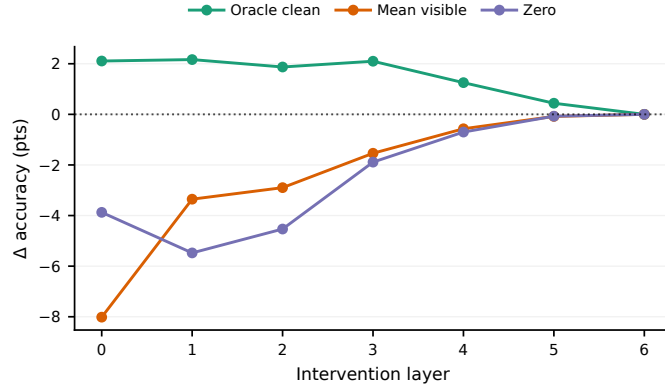


Figure A3: Causal replacement intervention for the Masked-Recon FT Plain ViT at 25% noise. Oracle clean-state replacement helps, while mean visible-token replacement and zeroing hurt.

Table A3: LSTM baseline hyperparameters for noisy-patch classification.

Hyperparameter	Value
Image size	32×32
Input channels	3
Patch size	4×4
Patch tokens	64
Blank compute steps	4
Patch embedding dimension	64
LSTM hidden dimension	128
LSTM layers	1
Learning rate	3×10^{-4}
Weight decay	0.05
Trainable parameters	104,138

Table A4: Plain and Rev ViT hyperparameters for noisy-patch classification.

Hyperparameter	Plain ViT	Rev ViT
Image size	32×32	32×32
Input channels	3	3
Patch size	4×4	4×4
Patch tokens	64	64
Class token	1	1
Patch embedding dimension	64	64
Position embedding dimension	32	32
Transformer input dimension	64	64
Transformer width	64	128 total, split as 64 + 64
Transformer layers	6	6
Attention heads	4	4
MLP ratio	4	4
Learning rate	3×10^{-4}	3×10^{-4}
Weight decay	0.05	0.05
Trainable parameters	312,170	321,258

Table A5: Hyperparameters for Masked-Recon and Clean-Recon ViT models.

Hyperparameter	Masked-Recon Plain	Masked-Recon Rev	Clean-Recon Plain	Clean-Recon Rev
Image size	32×32	32×32	32×32	32×32
Input channels	3	3	3	3
Patch size	4×4	4×4	4×4	4×4
Patch tokens	64	64	64	64
Class token	1	1	1	1
Patch embedding dim	64	64	64	64
Position embedding dim	32	32	32	32
Transformer input dim	64	64	64	64
Transformer width	64	128 total, 64 + 64	64	128 total, 64 + 64
Transformer layers	6	6	6	6
Attention heads	4	4	4	4
MLP ratio	4	4	4	4
Learning rate	3×10^{-4}	3×10^{-4}	3×10^{-4}	3×10^{-4}
Weight decay	0.05	0.05	0.05	0.05
Max epochs	300	300	500	500
Early stop patience	10	10	10	10
Train batch size	128	128	128	128
Trainable params	314,640	326,160	314,640	326,160

Table A6: Hyperparameters for the gated ViT experiments.

Hyperparameter	Value
Image size	32×32
Input channels	3
Patch size	4×4
Patch tokens	64
Class token	1
Patch embedding dim	64
Position embedding dim	32
Transformer input dim	64
Transformer width	64
Transformer layers	6
Attention heads	4
MLP ratio	4
Gate type	tokenwise
Gate update	convex combo
Gate init	0.5
Learning rate	3×10^{-4}
Weight decay	0.05
Max epochs	50
Early stop patience	8
Train batch size	128
Eval batch size	256
Trainable params	312,950